

Advisory Agent Walkthrough

Overview

The Advisory Agent has been successfully developed and integrated into the MVP. It acts as the final step in the inference pipeline, providing actionable, farmer-focused agricultural advice driven by the Google Gemini API, with a robust local fallback mechanism.

Changes Made

- **Environment & Dependencies:**
 - Installed `google-genai` and `python-dotenv` .
 - Exported the updated dependencies to `requirements.txt` .
- **Advisory Agent Module:**
 - Located at `[advisory_agent.py]`
(file:///d:/agricare/agents/advisory_agent/advisory_agent.py).
 - Integrates `google.genai` to dynamically request targeted advice based on disease class and severity.
 - Implements a seamless fallback knowledge base if the API is unreachable or the `GEMINI_API_KEY` is not present in the `.env` file.
 - Output strictly adheres to the requested JSON schema (Description, Symptoms, Treatment, Prevention, Fertilizer, Expert Consultation).
- **Streamlit Integration:**
 - `[1_Disease_Detection.py]`
(file:///d:/agricare/app/pages/1_Disease_Detection.py) updated to call the Advisory Agent.
 - Replaced the “Coming Soon” placeholders with dynamically rendered sections populated from the agent’s output. The advice now splits into a clean two-column layout showing symptoms/treatment on the left and prevention/fertilizer on the right.
- **Unit Testing:**
 - Written robust tests in `[test_advisory_agent.py]`
(file:///d:/agricare/tests/test_advisory_agent.py).

- Utilized `unittest.mock.patch` to mock out the `google.genai.Client` and simulate API responses without incurring network requests during local testing.

Validation Results

- Ran `venv\Scripts\python -m unittest tests/test_advisory_agent.py` .
- Ran 3 tests in 0.001s OK
- Verification passed:
 1. `test_fallback_logic` : Correctly returned hardcoded advice when no API key was present.
 2. `test_healthy_class` : Handled the edge case correctly without hitting the API.
 3. `test_gemini_mocked_call` : Processed a synthetic LLM response and parsed the custom schema exactly as expected.

The Advisory Agent is fully functional! If you add a `GEMINI_API_KEY=your_key` to a `.env` file in the root directory, the Streamlit app will immediately begin using Gemini. Otherwise, it will safely serve the fallback recommendations.